

数据库实验报告

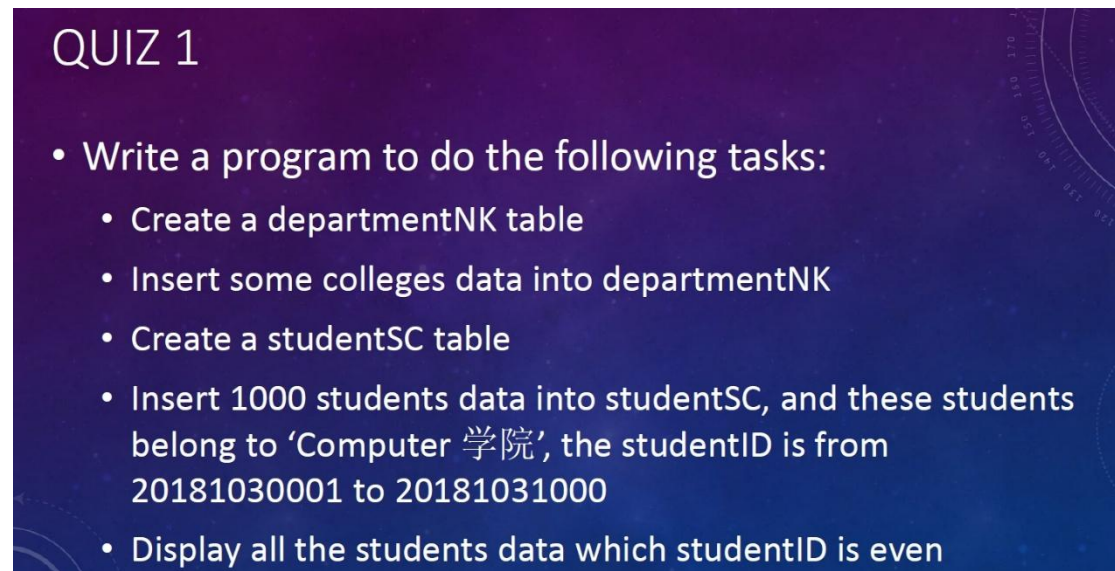
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实验题目：Python 与基础 SQL 语言

一、实验内容

使用 Python 实现如下图简单程序



QUIZ 1

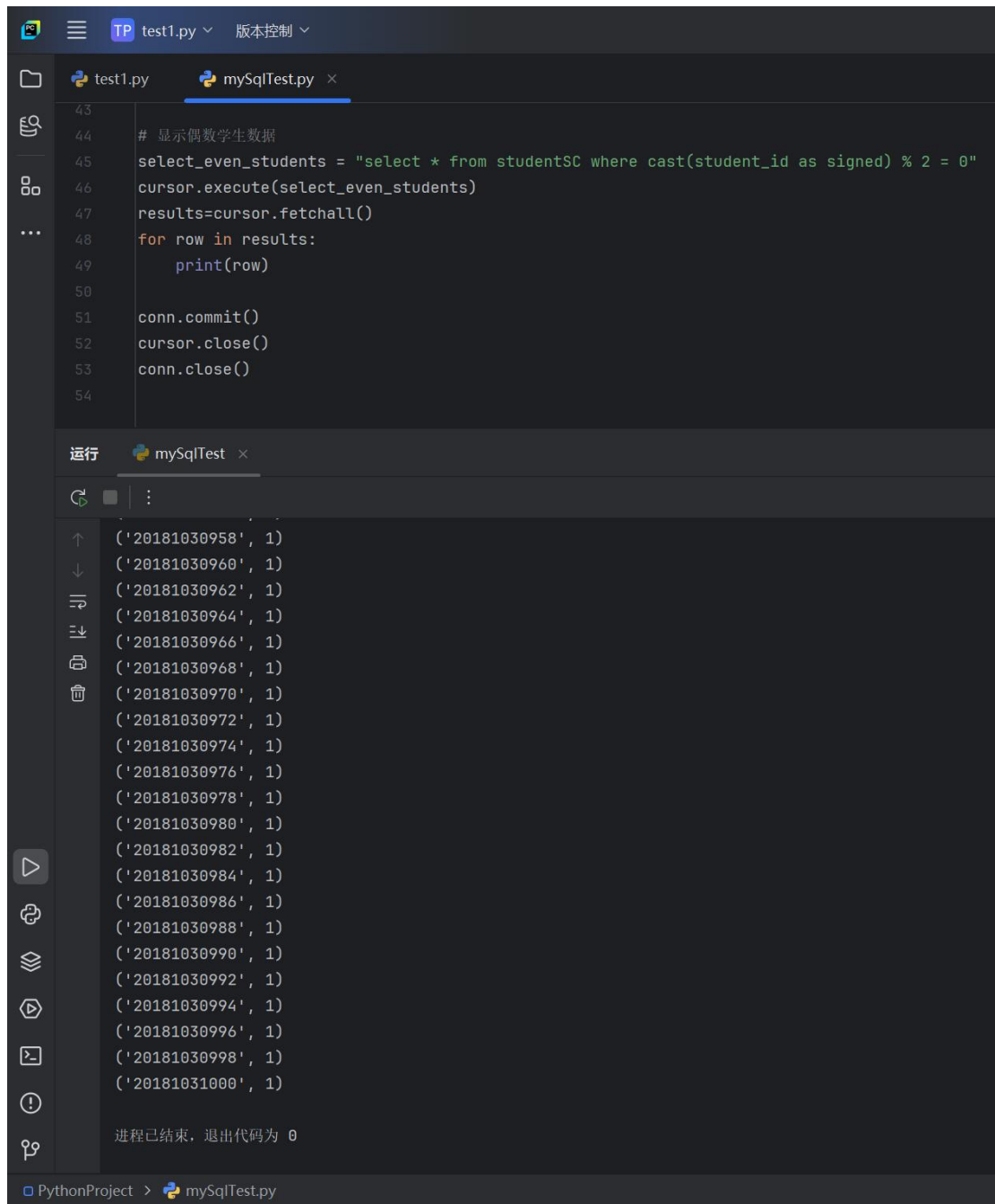
- Write a program to do the following tasks:
 - Create a departmentNK table
 - Insert some colleges data into departmentNK
 - Create a studentSC table
 - Insert 1000 students data into studentSC, and these students belong to 'Computer 学院', the studentID is from 20181030001 to 20181031000
 - Display all the students data which studentID is even

二、实现方法

```
1. import pymysql
2.
3. # 连接数据库
4. conn = pymysql.connect(host='localhost', user='newuserTest001',
5. password='Cmy20050611', database='dbscslab2018', charset='utf8')
6.
7. cursor = conn.cursor()
8.
9. # 创建学院表
10. create_department_table = """
11. create table if not exists departmentNK(
12.     college_id INT AUTO_INCREMENT PRIMARY KEY,
13.     college_name VARCHAR(20) NOT NULL
14. )
15. """
16. cursor.execute(create_department_table)
17.
18. # 插入学院数据
19. colleges = [('计算机学院',), ('人工智能学院',), ('软件学院',)]
20. insert_college = "insert into departmentNK (college_name) values (%s)"
21. cursor.executemany(insert_college, colleges)
22.
23. # 创建学生表
24. create_student_table = """
25. create table if not exists studentSC(
26.     student_id VARCHAR(20) PRIMARY KEY,
27.     college_id INT,
28.     FOREIGN KEY (college_id) references departmentNK (college_id)
29. )
30. """
31. cursor.execute(create_student_table)
32.
33. # 插入学生数据
34. cursor.execute(
35.     "select college_id from departmentNK where college_name = '计算机学院'"
36. )
37. result = cursor.fetchone()
38. college_id = result[0]
39. start_id = 20181030001
40. end_id = 20181031000
41. students=[(str(i),college_id)for i in range(start_id,end_id+1)]
42. insert_student="insert into studentSC (student_id, college_id) values (%s, %s)"
```

```
43. cursor.executemany(insert_student,students)
44.
45. # 显示偶数学生数据
46. select_even_students = "select * from studentSC where cast(student_id as signed)
    % 2 = 0"
47. cursor.execute(select_even_students)
48. results=cursor.fetchall()
49. for row in results:
50.     print(row)
51.
52. conn.commit()
53. cursor.close()
54. conn.close()
```

三、结果展示



The screenshot shows a Python IDE with two files: `test1.py` and `mySqlTest.py`. The `mySqlTest.py` file contains the following code:

```
43  
44 # 显示偶数学生数据  
45 select_even_students = "select * from studentSC where cast(student_id as signed) % 2 = 0"  
46 cursor.execute(select_even_students)  
47 results=cursor.fetchall()  
48 for row in results:  
49     print(row)  
50  
51 conn.commit()  
52 cursor.close()  
53 conn.close()  
54
```

The output window shows the results of the query, which are 20 rows of student data (student_id, score):

('20181030958', 1)
('20181030960', 1)
('20181030962', 1)
('20181030964', 1)
('20181030966', 1)
('20181030968', 1)
('20181030970', 1)
('20181030972', 1)
('20181030974', 1)
('20181030976', 1)
('20181030978', 1)
('20181030980', 1)
('20181030982', 1)
('20181030984', 1)
('20181030986', 1)
('20181030988', 1)
('20181030990', 1)
('20181030992', 1)
('20181030994', 1)
('20181030996', 1)
('20181030998', 1)
('20181031000', 1)

The output window also shows the message: 进程已结束，退出代码为 0.

四、结论

成功实现了实验目标，学会使用 Python 和 MySQL 语言相结合编写程序，熟悉了创建数据库、创建表以及插入数据等功能语法